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August 3, 2021

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To the office of Contracts and Grants:

On July 16, 2021, team String_teamName received a request for proposal commissioned by the University of Florida office of Sponsored Research, and jointly funded by the United States Department of Commerce and the United States Department of Energy. The request for proposal describes that they are seeking to endorse projects which "significantly improve the quality of life for individual American citizens at home, at work, or at school, or directly and immediately improve the quality and competitiveness of American industry within the global marketplace." Enclosed is team String_teamName's response. The proposed project, *OurCampus*, will improve the accessibility of student organizations, increase student safety, and serve as a valuable professional experience for computer science students.

The proposal is to facilitate the creation of an app by students, and for students. By doing so, students will be able to participate in a platform that aids in streamlined communication between the University, its student organizations, and the campus attendees. This endeavor will serve a threefold purpose. First, *OurCampus* will help foster the growth of student participation in on-campus organizations and the University community as a whole. Second, by streamlining the communication between the University and students, individuals will have increased access to important information and safety resources. Last, establish the *OurCampus* program will provide computer science students at the University of Florida with a unique opportunity to work on a legacy software engineering project as part of their educational experience.

We believe that the approval of the *OurCampus* app will provide a great deal of academic and social benefit for the University of Florida. Additionally, the app could serve as a template for future releases on other college campuses who are looking for ways to strengthen community involvement and promote student well-being in the digital age.

Sincerely,

Robert Dick, Connor Haley, and Andrew Sandell

OurCampus

Streamlining Campus Communication

Robert Dick, Connor Haley, Andrew Sandell

August 3, 2021

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Executive Summary

With the advent of social media, communication between students on a college campus has evolved to include a variety of online platforms. This shift has caused general communication between students, the university, and campus organizations to become diluted by the wealth of digital information that individuals must parse through in order to locate what they are looking for. At the University of Florida, students are in desperate need of a unified communications platform that allows for easy access to crucial information. The information most impacted by this proposal would relate to student organizations, campus navigation, and critical safety resources for students on campus.

To address this need, the University of Florida will fund and facilitate the development of a communication and content platform app for student use. The development of the project will be carried out over a number of semesters and will offer computer science students the opportunity to participate in a large-scale software engineering environment. The upkeep and maintenance of the application will be a legacy project, maintained by interested students each semester as an involvement opportunity through the CISE department. The app, named *OurCampus*, will feature organization matching algorithms, visual displays of room and event schedules, GPS pathing to classrooms, and night-time walk protection while on campus in the dark. Additionally, *OurCampus* will only be available to UF students and will be tied to each Gator's unique UFL email in order to prevent potential abuse of the application's community communication features.

By funding the development of the *OurCampus* app, the University of Florida will be fostering a close-knit campus community where students are more actively involved due to their ability easily find information related to their areas of interest. This will be of great benefit to the university as a whole, especially following the advent of Covid which has crippled many students experience of campus life. Additionally, by creating a comprehensive communication and information platform to unify the campus, students will have both increased access and awareness of university programs and resources that would be underutilized if not for the *OurCampus* app. While there have been many attempts at creating a unified social media for students, organizations, and administration on campuses, few have been able to effectively identify and achieve the target areas of weak communication. If *OurCampus* is successful in fostering a community built on efficient and accessible communication, it could be a highly marketable platform to other campuses.

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1. Problem Statement

The lynchpin of the college experience is effective communication. Whether that be through networking, attending events, or gaining valuable experiences from peers, involvement at the University of Florida is an immense opportunity for all who take the first step. Due to the lack of a centralized platform for communication, many students are out of the loop when it comes to getting involved on campus and don't know where to start their search, or where to find crucial safety resources for day-to-day use. The importance of student organizations to effective networking, growth, and resume building in the college years cannot be understated [7]. Without an accessible platform for communication between organizations and members, or for students to locate new organizations and event info, many students will go through college without ever getting involved in campus life[7]. While this may not impact many students' ability to participate in organizations, for many, the hurdle of information dispersal is the primary blocker to their ability to effectively plan for and follow through with attendance. Additionally, despite the massive increase in the usage of social media and mobile applications for communication, colleges have struggled to effectively adapt to the needs of its student population [6, 9]. Only recently has the University of Florida taken the first steps to increase safety awareness and resources related to sexual assaults on campus, and on a national level, students exhibit skepticism regarding their personal safety on college campuses [5, 4]. Ultimately, despite UF's efforts, it is clear that additional measures to ensure student safety are necessary to prevent future harm.

In order to address the need for unified campus communication among students, we propose the development of a social media app for UF students, by UF students. The app will be developed by students as part of a new involvement opportunity offered jointly through the UF

Administration and the CISE department. This opportunity will provide students with the opportunity to work on a meaningful project in an Agile development environment and gain experience in designing and interacting with content platforms in a professional and academic setting [10]. The app will feature a GPS map of the campus which students can use to temporarily post area-specific announcements and browse visual representations of organization meeting times and locations. To prevent abuse of the posting system, which has been a problem for similar campus-based applications [3], accounts will be tied directly to the user's UF email address. Additionally, the app will allow users to easily navigate organization pages with an intuitive matching and registration system. In order to answer the need of accessible digital resources for campus navigation, the app will allow users to generate pathing to specific rooms on campus–a popular feature of similar apps which has positively impacted many students and visitors with disabilities [1]. To answer the need for additional safety measures, the app's navigation system will feature a "Protect My Walk" mode which will notify UFPD if the student does not arrive at their destination.

2. Background and Research

2.1 Introduction

As the digital age has evolved over time, the wealth of information available to individuals has grown exponentially. Since the release of the original iPhone, the number of weekly app releases on the App Store has increased to 15,000 [6]. For many college students, this growth has only made communication more difficult, leaving many without a digital "home" on their campus despite the wealth of applications to choose from. As a result, college students in particular have become the target of many studies looking to assess the usage and effectiveness of social media in an academic setting [10]. The deluge of orientation information, needless advertisements, and personal contacts obfuscates important student organizations, exciting events, and important activities. Despite this hurdle, students primarily share news and events related to their local community as opposed to international events or politics [9]. This indicates that students oftentimes feel a strong connection to their university's locale and consider themselves community members. However, online advertisements and social media marketing are far and away, one of the most effective ways to promote and share event information [8]. Thus, it is important that whatever solution is implemented harnesses the power and efficiency of social media rather than attempt to subvert it. An additional problem is that students both new and old oftentimes struggle to navigate their way around the campus due to poor online documentation, unannounced construction, and lack of familiarity with the campus. During night-time, this problem is only exacerbated by the lack of substantial safety resources by the University to guard its students. While UF has taken slight steps towards addressing the accessibility of student resources, there remains a significant need for a centralized, comprehensive source of student and organizational information that is available in an easy-to-use online format [5].

2.1.1 Related Literature

There have been numerous attempts to create a unified hub for communication between students, organizations, and faculty at other college campuses in the past several years, with each falling short in one aspect or another. *Access My Campus* was a mobile phone application that was being developed in 2014 for the purposes of streamlining campus navigation [1]. Users would be directed by GPS to the exact classroom they desired, while offering alternative paths along the way. Additionally, *Access My Campus* was particularly designed to accommodate students with disabilities that hindered day-to-day navigation [1]. Another app designed for students to use in daily campus life was *Yik Yak* [3]. *Yik Yak* launched as a fully anonymous community exclusive to students attending their college [3]. Users were able to post announcements, coordinate with other members, and upload media throughout the day. Unfortunately, this app suffered from a lack of moderation among many communities, which lead to an early downfall in the public eye and a sharp decrease in active users [3]. Aside from centralizing campus communications and navigations, a recurring concern of college students is the safety of college campuses after dark or in low traffic areas [4] and the availability of online resources to help protect students [5].

The purpose of this study is to assess student attitudes regarding the centralization of campus communication, community social media, and critical student safety resources into a single digital platform. The app, dubbed *OurCampus*, would offer students the ability to post organization announcements, connect and match with recommended organizations, view live maps of club room reservations on campus, generate pathing to desired locations on campus, and implement a temporary tracking system to help protect students while walking along at night. In order to determine student's attitudes toward such a solution, sentiments regarding campus involvement, safety, and accessibility of information had to first be assessed.

2.2 Methodology

As part of the research process, team string_teamName elected to conduct a study using UF students as a population sample. In order to determine student needs, the team agreed that the survey needed to effectively assess student's attitudes related to organizational involvement, campus communication, and personal safety at the university. Additionally, the team also needed to make sure that the collected data could provide an accurate reflection of each student's desire for a solution and how much each respondent is concerned with the posed issue. Ultimately, a nine-question survey was created and approved by the team to accomplish these goals.

2.2.1 Materials

The chosen method of data collection was a Qualtrics survey, comprised of nine short questions. Qualtrics was chosen as the survey platform due to its easily accessible results, filter query functionality, and automatic data graphing that is updated in real time. These features provided valuable information that otherwise would have been difficult to clean, balance, and graph appropriately. Additionally, Qualtrics was judged to be a suitable platform due to UF student's familiarity with the site and its functionality.

2.2.2 Procedure

First, the team discussed the target data that needed to be collected to draw meaningful conclusions from the survey. Second, these target areas of data were translated into actionable questions, ranging from simple ranked responses to involvement percentage estimates on a slider bar. After construction was completed, the Qualtrics survey was sent out to several student organizations and posted on the ENC3246 discussion board on July 19th and was available for a week of data collection. By July 27, the survey had amassed a total of 21 anonymous respondents. While more responses would have been preferred, given the circumstances of both Covid-19 and a shortened period of data collection, the lower number was anticipated. Despite this, the results still provide clear and actionable data for the team to use.

2.2.3 Survey Breakdown

As previously mentioned, the survey was designed to address the three primary areas of difficulty that the team is interested in solving with *OurCampus*. These areas are−student involvement and communication within the campus community, tools to improve student navigation, and resources to better protect UF students who feel at risk while traversing campus. The form asks students how well they know what activities and events are available on campus, and if they know how to access information related to campus involvement. Additionally, questions related to each respondent's current level of involvement and ideal level of involvement are collected in order to better understand what type of outcome respondents would ideally desire. The survey concludes with questions related to implementing safety procedures on campus at night, and how confident each respondent is in their personal safety on campus at the moment.

2.3 Results

Our first question asked respondents if they know what student events are happening on campus on any given night. Responses were limited to five selections, ranging from "Definitely not" to "Definitely yes." Responses indicated overwhelmingly that 86% of polled students were not confident in their knowledge of what events or student meetings were taking place on campus, with no respondents indicating moderate or extreme confidence in their ability to do so (Figure 1).

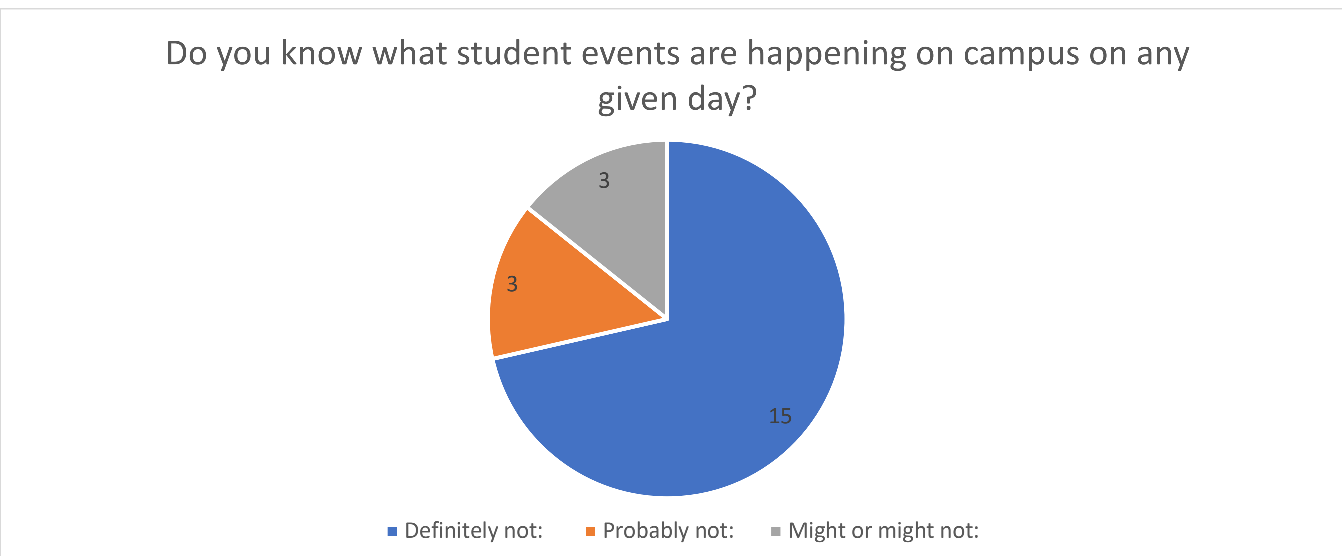


Figure 1 - Confidence of Respondents' Knowledge of Student Events on Campus

The results of the survey also indicate that 81% of students were not confident in their ability to locate relevant information related to student organization meeting times, scheduled events, and other activities on campus (Figure 2)

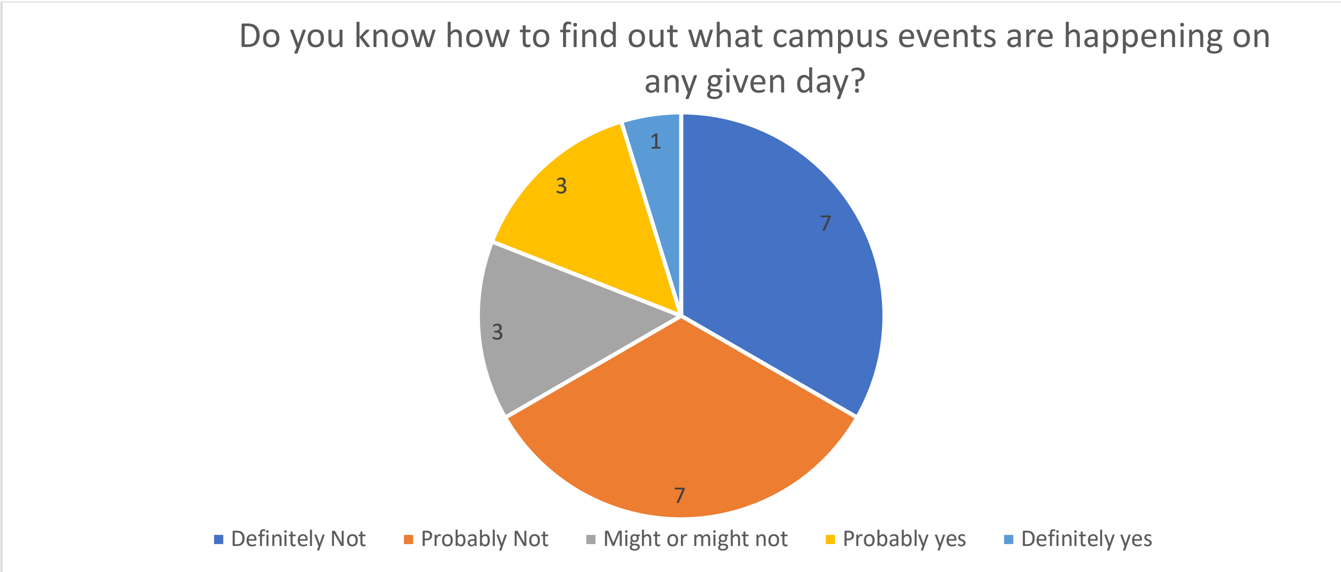


Figure 2 - Confidence of Respondents' Knowledge of How to Locate Event Information

The survey also assessed student's self-report level of participation in campus activities and organizations on a scale of 1 – 100 (Figure 3). The results of this question were displayed using a histogram which indicates the respondent's reported percentage.

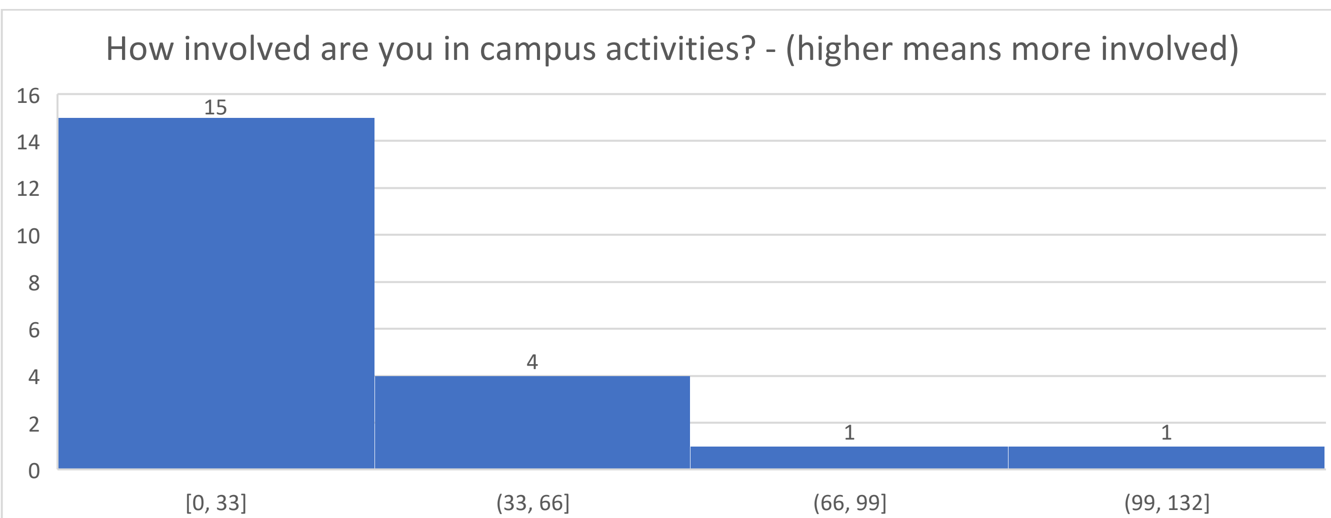


Figure 3 - Respondents' Reported Involvement in Campus Activities

Additionally, respondents provided their desired level of participation in campus activities and organizations on the same scale as the previous question (Figure 4). The results of this question were once again displayed using a histogram which indicates the respondent's reported percentage.

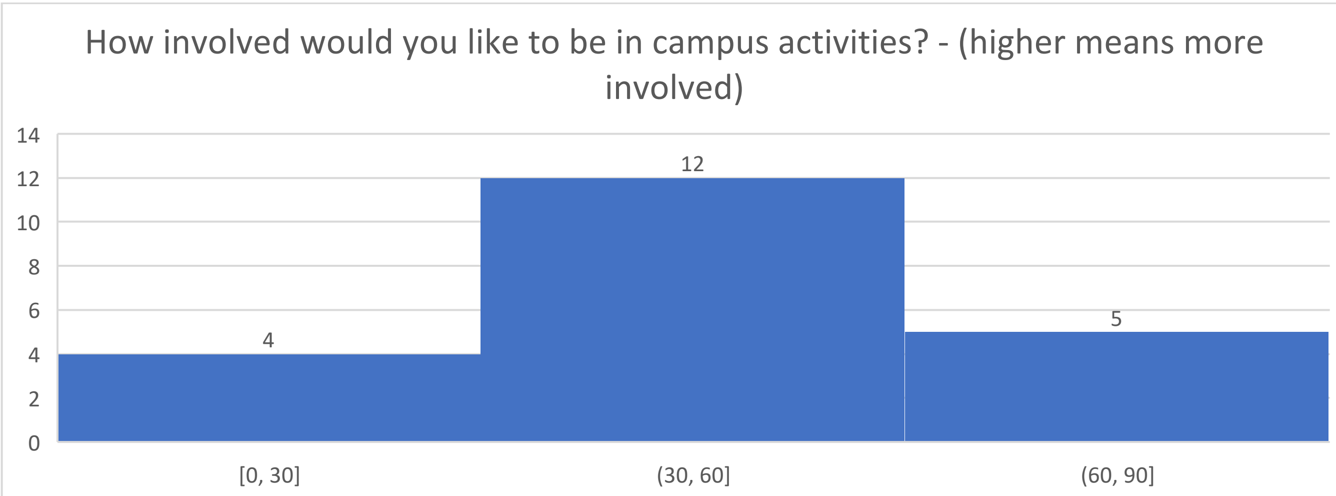


Figure 4 - Respondents' Reported Desired Involvement in Campus Activities

From this data, the team was able to construct the "Desired Involvement Growth Index" which represents each respondent's net desired involvement growth from their current level of involvement (Figure 5). In short, positive values indicate a respondent wishes to become more active in campus activities, negative values indicate a respondent wishes to become less active in campus activities. Overall, the data suggests a positive average for desired involvement.

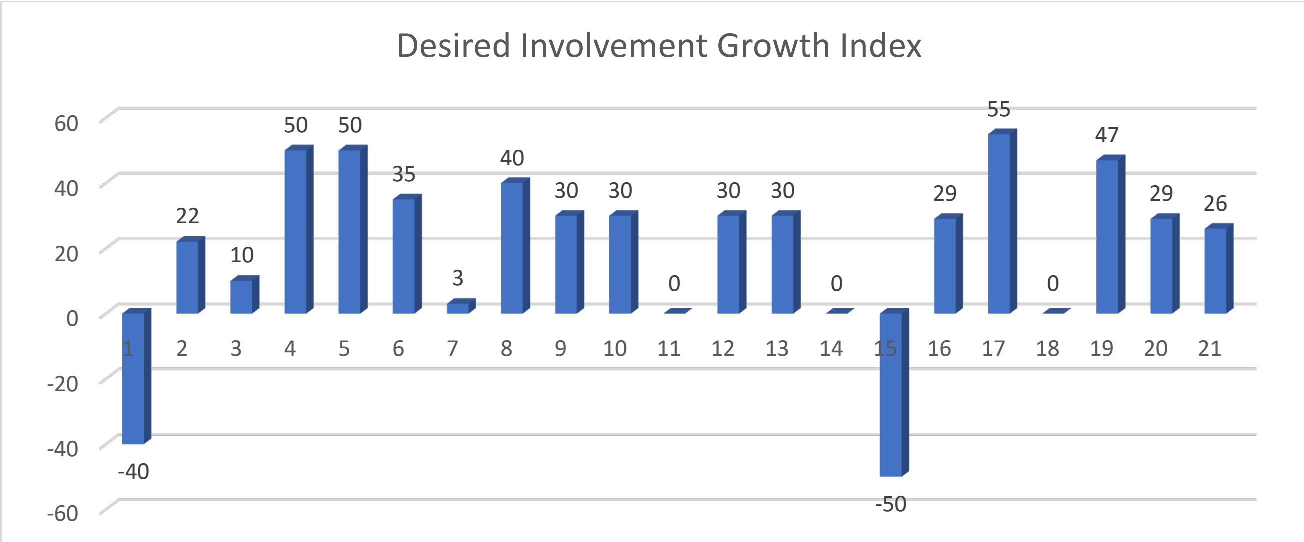


Figure 5 - Desired Involvement Growth Index of Respondents

Next, the survey asked respondents to indicate how safe they feel on campus at night by choosing one of the five options ranging from "Definitely not" to "Definitely yes" (Figure 6).

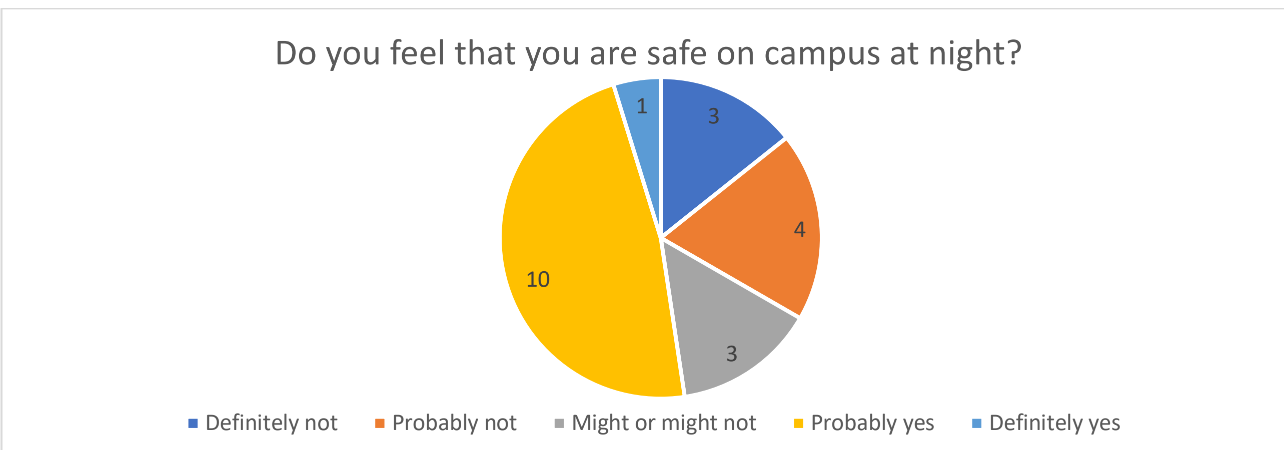


Figure 6 - Respondents' Sentiments Towards Their Personal Safety on Campus at Night

Following from this response, respondents were asked whether they felt that additional safety measure should be implemented to ensure campus safety at night. Once again, option choices ranged from "Definitely not" to "Definitely yes" (Figure 7). The data indicates that 90% of respondents believed that additional safety measures are necessary to protect students on campus at night.

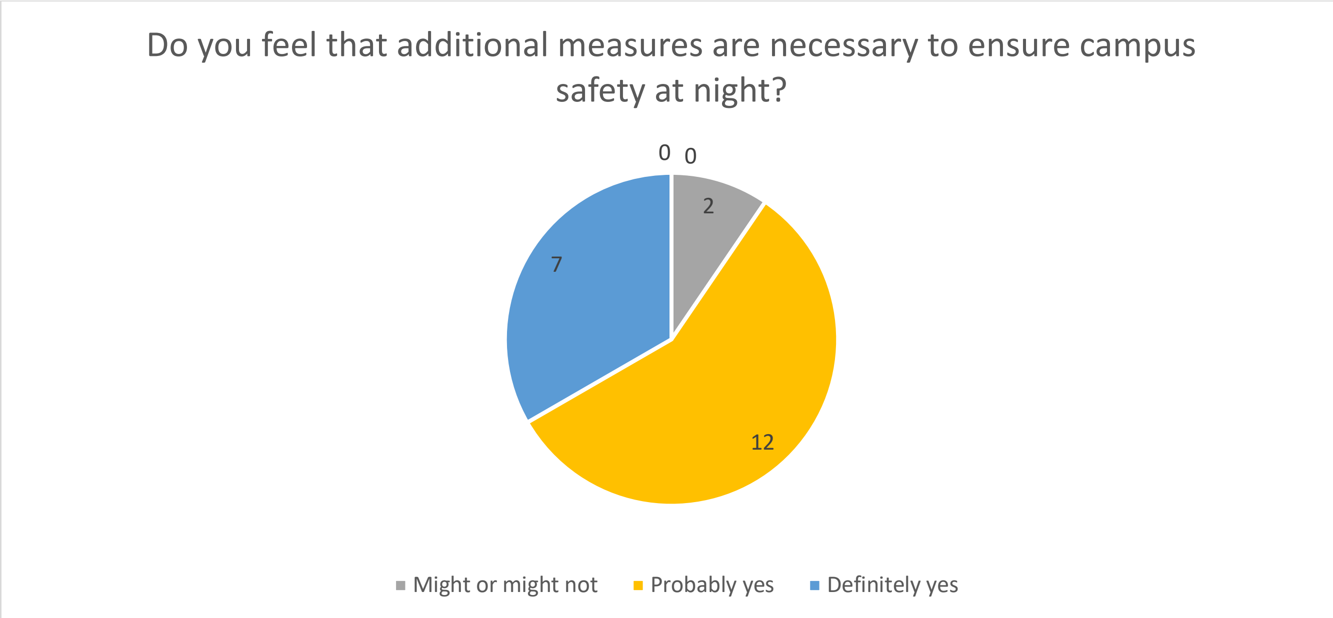


Figure 7 - Respondents' Attitudes Towards Implementing Additional Safety Measures on Campus

Lastly, the team measured the responses to perceptions of safety on campus alongside the reported gender of each respondent (Figure 8). This chart helps to further separate the data to see if there are any trends alongside reported gender. Columns labeled “(M)” indicate a male respondent, “(F)” indicates a female respondent, and “(N)” indicates a non-binary or other respondent.

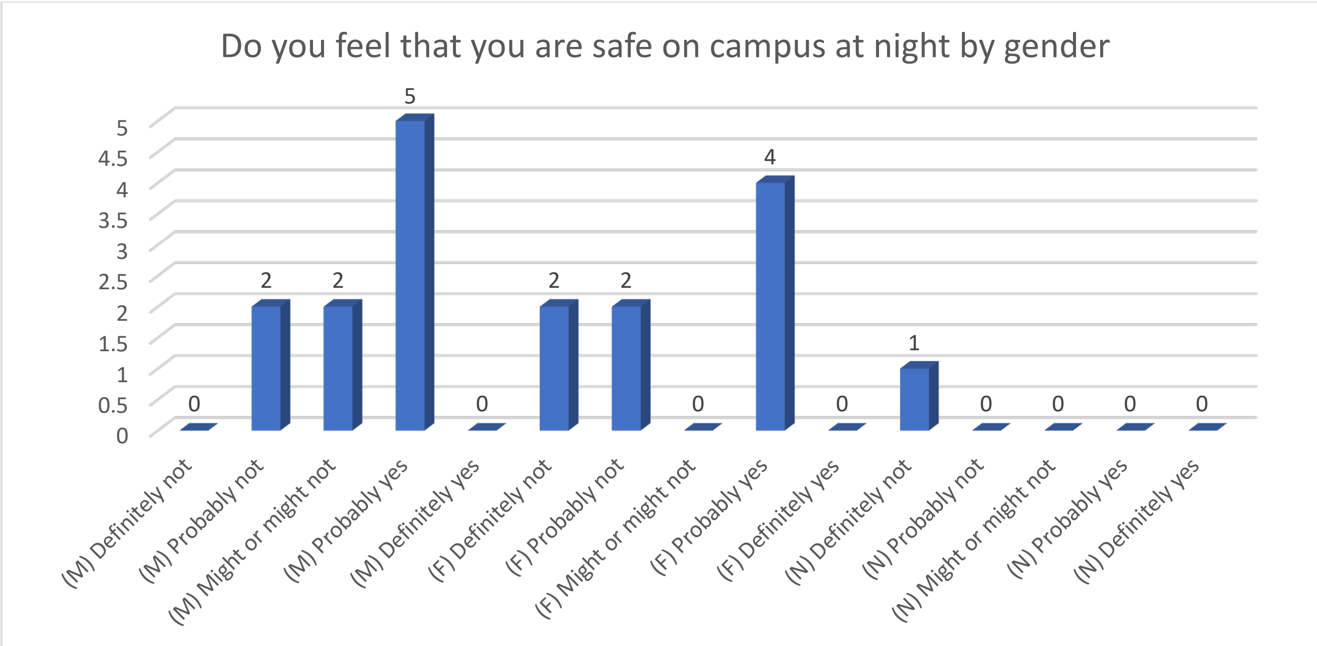


Figure 8 - Respondent Perceptions of Safety by Gender

2.4 Discussion

2.4.1 Analysis

From the responses to the survey, the team was able to gather some crucial information regarding student attitudes towards issues present with university communications. Figures 1 and 2 indicate that the vast majority of students do not stay up to date with campus involvement information, and more importantly, they do not know where to find that information in the first place. This is important to keep in mind, especially when considering the responses illustrated in figures 3, 4, and 5. Figure 3 displays the self-reported level of involvement that students reported, with 71% of respondents indicating that they are barely, if at all involved in on campus activities. Figure 4 on the other hand indicates each respondent's desired level of participation in on campus events, with 19 % of respondents indicating that they wish to remain barely involved, if at all, 57% wanting to be moderately involved, and 24% wanting to be very involved. This indicates an immense trend in student sentiment towards a growth in personal involvement in campus activities, which is further reflected in figure 5.

Figures 6, 7, and 8 address the team's desire to assess student's perceptions of personal safety. In figure 6, 48% of respondents held a negative or mixed certainty in their safety on campus at night, with the rest of respondents feeling somewhat certain of their personal safety. This prompted the creation of figure 7, which illustrates student sentiments towards implementing additional safety measures to protect students on campus during the night. 90% of students were strongly in favor of implementing these changes.

From these responses, the team is confident that a mobile app similar to *OurCampus* will address the indicated student needs. *OurCampus*'s organization recommendation system, announcement posting, and scheduling map will allow students to easily discover and plant to attend new organizations, UF events, and stay in sync with the heartbeat of the UF community. Additionally, *OurCampus*'s "Protect my Walk" mode will offer students the much needed and desired protection as they make their late-night walks to and from the library or across campus. However, with this feature, extra precautions must be taken to ensure the user's data privacy. As a result, it will be necessary to collect the user's consent to location sharing in order to remain in accordance with typical data privacy regulations [2].

2.4.2 Limitations

Due to the condensed timeframe the survey was conducted under, as well as the ongoing Covid-19 pandemic, data gathered from responses may reflect an inaccurate view of student sentiment upon return to campus. Many students have not had the chance to get involved with the UF community, so it is important to account for the possible over-enthusiasm or under-reporting of involvement.

2.4.3 Conclusion

The data collected from the survey indicates that students at the University of Florida are incredibly eager to increase their involvement on campus, but struggle to find information on where to start. Additionally, while many students feel moderately safe navigating campus at night, a large proportion are much less confident in their ability to do so. Overwhelmingly, students are in support of additional safety resources to help protect them during their time on campus. This data is valuable to the team because it demonstrates that the features offered by *OurCampus* are valuable to the UF community and would be widely accepted as solutions to the problems faced by students on a daily basis.

3. Technical Plan

The team proposes to create a mobile application that students can install on their Android or IOS personal devices. We believe that this approach is best suited for the college student because GPS functionality is essential for tracking the user's location, which will be used in certain features such as walk protection and listing student event locations. In the field of software engineering, the Software Development Life Cycle (SDLC) is widely recognized and utilized by various organizations to produce quality software [11]. This section will outline the SDLC phases of Requirement Definition, Design, Development, Testing, and Maintenance as they relate to the creation of this mobile application. Some sources list additional phases in the SDLC, but those phases can generally fit within these five.

3.1 Requirement Definition

There are several requirements for the *OurCampus* mobile application. The mobile application should be used by as many students and faculty as possible, so it should be available for both Android and IOS devices. The application needs to utilize a Global Positioning System (GPS) to track user and event locations. A central database is essential, and it can be communicated with through an Application Programming Interface (API). Users will need to be verified to be students or faculty, and an administrator panel web user interface will be necessary to perform actions such as restricting user accounts and providing information to law enforcement. More requirements may be identified in the future.

To list the predetermined requirements:

- Cross-Platform compatibility
- GPS
- Administrative panel web application
- API to link the application and the database
- Database
- User verification

3.2 Design

This section will give preliminary ideas for how the mobile application can be designed, such as what software and third-party libraries might be used.

3.2.1 Cross-Platform compatibility

Because the application needs to be used by nearly the entire university, it needs to be available for both IOS and android devices. To achieve this, the applications will be created with React Native. This allows a very large percentage of the codebase to be reused across different operating systems. As an example, the engineering team from Nerdwallet reported that approximately 95% of the mobile app code was reused across operating systems thanks to React Native [12].

3.2.2 GPS

Because the application needs to have a reliable way to track user, event, and building locations, one possible solution is using the Google Maps API. Significant work has already been done by an engineering team at Google, so it is most logical to use their existing APIs for maps and routes than start from scratch.

3.2.3 Administrative Panel Web Application

To perform administrative actions, limited faculty members should be provided with access to an administrator panel web application. This web application will provide them with the capability to perform actions on user accounts, modify or delete events, and check user location data for the safe walk feature. These functions are essential to prevent spam, correct user errors, and cooperate with law enforcement officials if a student uses the safe walk feature and an incident happens during the time of their walk. Essentially, this web application provides a way for administrators to interact with the database in a less restricted yet user-friendly way.

3.2.4 API

To share user and event information, the mobile application will need to communicate with an API. This will be the back end that creates a bridge between the mobile application (or web application) and the database, while restricting user actions by their permissions.

3.2.5 Database

This will probably be in the form of one central database with multiple tables used to store data about events and users. There are many available database servers; one that may be used is the MySQL server.

3.2.6 User verification

The back-end server that contains the API will need to have a way to send emails to verify that a student or faculty member owns a UF email address. The user will click on the link in their email address to have their email address verified. It is important that the email address that this is sent from is not blocked by UF email security policies and it is preferable that a UF email address is used to send these verification emails.

3.2.7 Additional Design Requirements

Additional design requirements will be specified by the engineering team responsible for creating the mobile application, as they relate to more specific items such as the user interface or hardware.

3.3 Development:

The engineering team will proceed to develop the necessary components for the *OurCampus* application. This can be done by creating multiple GitHub repositories, in which the code for each component will be stored and can be modified. Git is a method of version control, which is implemented through GitHub. Each developer can push their changes to the relevant GitHub repository, which has many benefits, the most important of which is the ability for multiple developers to make changes to the same file at the same time without conflicts [13].

During the development process, static code analysis tools will be used to identify flaws and correct them before they can be pushed to an environment. This is important to reduce the number of security vulnerabilities introduced into the software. The process of integrating security into all stages of development is known as DevSecOps [14].

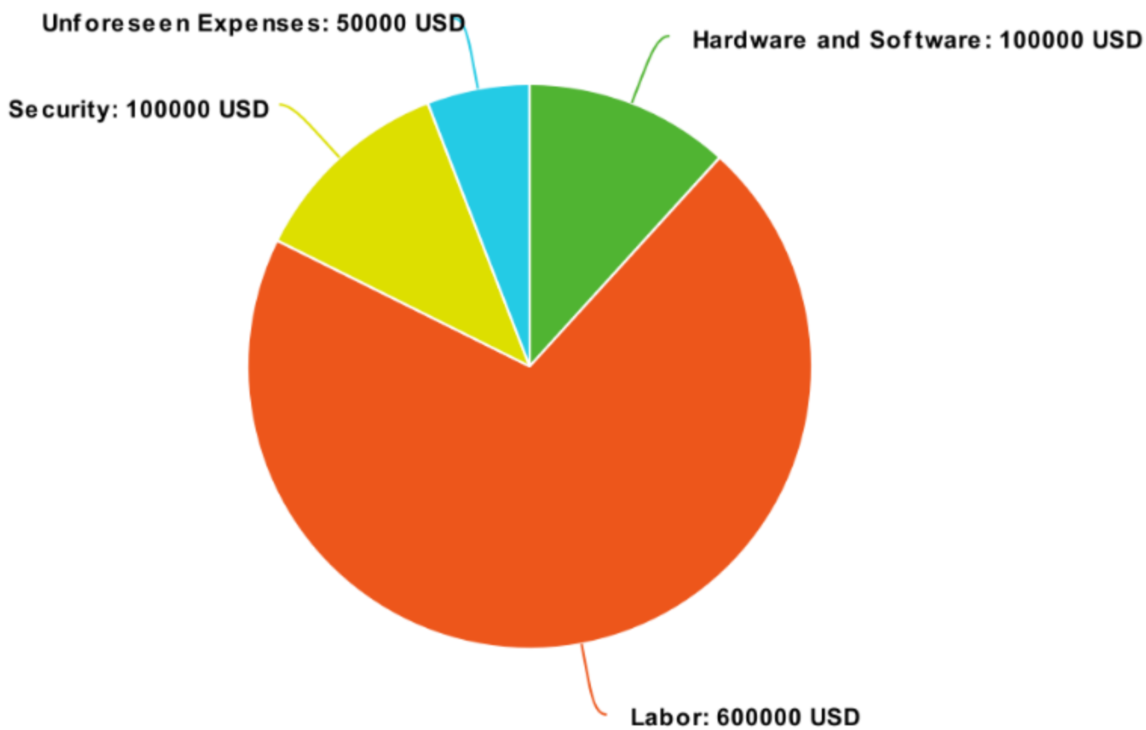
3.4 Testing:

When developers create software, they test their software within multiple environments. These environments typically consist of the development, staging, and production environments. Each environment is essentially a clone of the other, with the possibility for slight changes. The development is used frequently by developers as they create the software. The staging environment is a testing environment for finished software to perform quality assurance. The production environment is the final environment that end users interact with. Software must first be pushed to the development and staging environments before the production environment to prevent disrupting the user experience [15].

In addition to traditional quality assurance testing, the applications should undergo penetration testing to ensure the safety of user data. Penetration testing is the process of identifying vulnerabilities to assess risk. An ethical hacker will look for vulnerabilities in systems, exploit them, and report them. This penetration testing should be performed by a third party as hiring staff will prove to be significantly more costly [16].

4. Budget and Schedule

As the SDLC is a continuous process, the budget may change over time. To begin, we recommend distributing the funds as so:



As can be seen from the chart, we anticipate the largest expense to be labor, as developing the mobile application, API, and administrative panel will be the costliest. In addition, labor is required to design the database, prepare the various environments, and design the user interfaces. We expect that these labor costs will cover the initial development of the *OurCampus* software, and that labor costs will be reduced after the release because less engineers will be required to maintain the project. There is a section of the chart dedicated to unforeseen expenses, as it is unlikely that every expense can be accounted for in this plan.

In our proposed schedule, the initial development of the application will last no more than six months. In the first week, a complete plan for development will be created. Simultaneously until the end of the first month, engineers will create the necessary database and approximately half of the API functionality. Then, until the end of the second month, engineers will develop the administrative panel and finish developing the API functionality. In the third and fourth months, engineers will develop the main mobile applications for IOS and android. In the fifth month, thorough quality assurance and penetration testing will be completed. In the sixth month, final changes to the software will be made and tested. Then, the mobile application will be released to the public slowly, with limited marketing so that system flaws can be identified and fixed. As the team becomes more confident in the quality of the software, the university can encourage more users to install the mobile applications. Ongoing maintenance and security testing will be performed.

5. Evaluation Plan

Each month after release, the project's success will be evaluated through analytics data and surveys. The analytics data will provide information on student and faculty usage, such as the number of concurrent users, total users, and commonly used features. The surveys will provide information on the overall popularity of the mobile applications on campus, specifically student and faculty opinions, which the analytics data will lack. This feedback will allow the developers to make changes to the software with the goal of improving the user experience and increasing the overall usage of the mobile applications.

Six months after the release, we will evaluate the success of the mobile applications primarily from two metrics: the number of unique users, and the average rating given by users. We will view the project as a success if at least one third of all students attending the University of Florida have installed the mobile application and the average rating is at least 6/10 according to the most recent survey. If the targets are not met, the project will be considered unsuccessful, and a decision will be made whether to discontinue the project or make more significant improvements and design changes. If the project is overwhelmingly successful, a decision will be made whether to expand to other universities. As the software and project will be owned by the University of Florida, these decisions would be made by the appropriate university faculty members. In addition, the University of Florida will have a financial incentive to expand to other universities through licensing agreements or a Software-as-a-Service (SaaS) business.

6. Appendix

Survey Questionnaire:

1.) Do you know what student events are happening on campus on any given day?

Definitely Not | Probably Not | Might or Might Not | Probably Yes | Definitely Yes

2.) Do you know how to find out what campus events are happening on any given day?

Definitely Not | Probably Not | Might or Might Not | Probably Yes | Definitely Yes

3.) How involved are you with campus activities?

Scale of 0 – 100

4.) How involved would you like to be in campus activities?

Scale of 0 – 100

5.) Do you feel that you are safe on campus at night?

Definitely Not | Probably Not | Might or Might Not | Probably Yes | Definitely Yes

6.) Do you feel that additional measures are necessary to ensure campus safety at night?

Definitely Not | Probably Not | Might or Might Not | Probably Yes | Definitely Yes

7.) What is your gender?

Male | Female | Non-Binary/Third Gender | Prefer Not To Say

8.) Do you attend the University of Florida?

Yes | No

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